HOCHSCHULE FULDA
UNIVERSITY OF APPLIED SCIENCES

Project Management Abstract & Technical Specification Document.

Course: Machine Learning for Engineering Applications.

Document: Project Management Abstract & Technical Specification (Deliverable 1)

Project Title : Simulation-Based Design Space Optimization Using Asynchronous Advantage Actor–Critic (A3C)

Team Members.

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Submission Date: 29 January 2026.

Version History.

The following version history documents the iterative development, review, and refinement of the *Project Management Abstract & Technical Specification* throughout the project life cycle. Each version reflects incremental improvements in scope definition, technical content, requirements engineering, and quality assurance activities.

Version	Date	Description of Changes	Author(s)	Reviewer(s)
0.1	17.01.2026	Initial document structure created; project scope and objectives outlined.	Shariqa Moin, Rashid Hussain	Revathi Podaralla
0.2	18.01.2026	Preliminary task list, role model, and initial RACI matrix defined.	Rashid Hussain	Shariqa Moin
0.3	19.01.2026	Role descriptions refined; non-essential roles removed to match PoC scope.	Revathi Podaralla	Rashid Hussain
0.4	21.01.2026	Project description updated to reflect simulation-based A3C optimization.	Shariqa Moin	Revathi Podaralla
0.5	22.01.2026	Timeline and milestone definitions aligned with actual project execution.	Rashid Hussain, Revathi Podaralla	Shariqa Moin
0.6	24.01.2026	Technical specification expanded with Cockburn use cases and PARIS requirements.	Rashid Hussain	Shariqa Moin, Revathi Podaralla
0.7	25.01.2026	UML activity diagram added to model A3C control flow and simulator integration.	Revathi Podaralla	Rashid Hussain
0.8	26.01.2026	Cross-section consistency review (requirements, diagrams, roles, milestones).	Shariqa Moin	Project Team
1.0	27.01.2026	Final formatting, quality check, and submission preparation.	Project Team	Project Team

Revision Highlights

- Progressive refinement from initial project scope definition to a finalized management and technical specification document.
- Alignment of roles, milestones, and requirements with the selected A3C-based simulation optimization approach.
- Incremental addition of UML activity diagrams, use cases, and formal requirements to ensure technical completeness.

Project Description.

This project focuses on the simulation-based optimization of engineering design parameters using a reinforcement learning approach, specifically the Asynchronous Advantage Actor–Critic (A3C) algorithm. The objective is to explore and optimize a multidimensional design space by iteratively interacting with an external engineering simulation model.

The simulation model, provided as the executable ConsumptionCar.exe, evaluates candidate design configurations and returns performance-related metrics. These metrics are used to guide the learning process of the A3C algorithm toward high-quality design solutions.

The project is implemented as a proof-of-concept design space exploration (DSE) system, demonstrating how modern reinforcement learning techniques can be applied to engineering optimization problems where analytical models are unavailable or impractical.

Motivation and Relevance.

Engineering design problems often involve:

- Nonlinear relationships
- Black-box simulations
- Conflicting objectives
- Expensive evaluation functions

Traditional optimization techniques struggle in such settings, especially when gradients are unavailable. Reinforcement learning offers an alternative by learning directly from interaction with the simulation environment.

The A3C algorithm is particularly suitable due to:

- Its ability to run multiple parallel workers
- Efficient exploration of the design space
- Scalability for computationally expensive simulations

This project demonstrates how reinforcement learning can complement or replace classical optimization approaches in simulation-driven engineering workflows.

System Concept and Workflow

The developed system consists of three main components:

1. Reinforcement Learning Optimizer (A3C)
   Generates candidate design parameter sets and updates the optimization policy.
2. Simulation Environment (ConsumptionCar.exe)
   Evaluates each design configuration and returns performance metrics.
3. Evaluation and Analysis Layer
   Aggregates results, computes rewards, and visualizes optimization progress.

Each optimization iteration is treated as a single-step episode, which is a suitable abstraction for design space exploration where each design evaluation is independent.

Input Data and Design Parameters

Optimization Inputs

The primary inputs to the optimization process are five continuous design parameters, generated by the A3C actor network in normalized form [0,1]. These parameters are subsequently mapped to real engineering values before being passed to the simulation.

Parameter	Description	Lower Bound	Upper Bound
P1	Engine-related design parameter	3.00	5.50
P2	Secondary system parameter	0.40	0.50
P3	Gear ratio (G3)	0.50	2.25
P4	Gear ratio (G4)	0.50	2.25
P5	Gear ratio (G5)	0.50	2.25

Constraints

To ensure physically meaningful designs, the following constraint is enforced before simulation execution:

- Gear hierarchy constraint:

$$ig_3 > ig_4 > ig_5.$$

If a sampled action violates this constraint, it is corrected prior to simulation execution.

Simulation Interface and Data Exchange

Data Sent to the Simulation

For each episode, the optimized real-valued design parameters are passed as command-line arguments to the external simulation executable:

```
ConsumptionCar.exe <P1> <P2> <P3> <P4> <P5>
```

The simulation is executed as a subprocess, allowing seamless integration into the optimization loop.

Simulation Outputs

The simulation returns multiple performance indicators via standard output. The relevant outputs used in this project are:

Output	Description
fc	Fuel consumption
el_g3	Elasticity of gear 3
el_g4	Elasticity of gear 4
el_g5	Elasticity of gear 5

These values form the basis for evaluating the quality of each design configuration.

Reward Formulation and Optimization Objective

The optimization problem is formulated as a scalar reward maximization task that balances multiple engineering objectives.

The reward function is defined as:

$r=(elg3+elg4+elg5)-2\cdot fc$

Where:

- Fuel consumption is penalized
- Elasticity metrics are rewarded

This formulation implicitly represents a multi-objective trade-off, guiding the optimizer toward solutions with low fuel consumption and high system elasticity.

Optimization Outputs and Results

The primary outputs of the system include:

- Optimized design parameter sets
- Reward values for each evaluated design
- Simulation performance metrics
- Best-performing design configuration
- Visualization of optimization progress

The results are stored and visualized through:

- Reward progression plots
- Parameter evolution plots
- Simulation output plots

These outputs enable both quantitative evaluation and qualitative interpretation of the optimization process.

Expected Outcome

The expected outcome of this project is:

- A functioning A3C-based optimization prototype
- Successful integration with an external engineering simulation
- Demonstration of parallel, simulation-driven design space exploration
- Identification of high-performing design configurations

The project serves as a foundation for extending reinforcement learning approaches to more complex engineering optimization tasks.

Roles & Responsibilities (RACI-Matrix).

To ensure a structured and efficient execution of the project, roles and responsibilities were clearly defined using a RACI (Responsible, Accountable, Consulted, Informed) matrix. The RACI model provides transparency regarding task ownership, decision authority, and collaboration between team members. Given the interdisciplinary nature of the project, responsibilities were distributed to balance project coordination, machine learning development, and simulation-based evaluation activities.

Tasks & Competencies for AI-Based Systems	Project Manager	Requirements Engineer	Integration Engineer	Test & Validation Engineer	Data Engineer	Machine Learning Engineer
Integration of the A3C reinforcement learning algorithm into the simulation-based optimization software architecture	I	I	C	I	C	R
Selection and justification of the A3C reinforcement learning algorithm according to system requirements	I	C	I	I	C	R
Definition of the multi-objective optimization problem including reward formulation and system constraints	I	C	I	I	C	R
Definition of simulation input/output handling and reward data management for A3C training	I	C	C	C	R	C
Definition of quality criteria for simulation outputs and reward signals used during A3C training	I	C	I	I	R	C
Configure A3C hyper parameters and actor–critic network architecture	I	C	I	I	C	R
Execute A3C training using parallel workers and simulation interaction	I	I	I	I	C	R
Analyze reward convergence and identify high-performing design configurations	I	C	I	C	I	R
Evaluation of A3C optimization results based on defined acceptance criteria and system requirements	C	A	I	R	I	C

Role Model Justification: The selected roles are derived from the professional AI system development setup presented in the course material and tailored to the scope of this project. For this specific project, the team focuses exclusively on the roles that are essential for the development of a conceptual proof-of-concept for simulation-based design space exploration using the Asynchronous Advantage Actor-Critic (A3C) reinforcement learning algorithm.

The chosen roles cover project coordination, requirements definition, machine learning algorithm design, simulation data handling, system integration, and result validation. Roles related to deployment, legal compliance, ethics, and long-term maintenance were deliberately excluded, as the project is limited to a closed, offline academic prototype using a provided simulation model without real user data or operational deployment.

This focused role selection ensures that all relevant engineering, data, and management responsibilities are addressed in a coherent and efficient manner by the three team members while maintaining consistency with the project scope and the defined technical objectives.

Role Description.

Team Member	Assigned Roles	Core Responsibilities
Rashid	• Machine Learning Engineer(A3C) • Data Scientist • Systems Engineer	Responsible for defining the multi-objective optimization problem including reward formulation and system constraints, designing and configuring the Asynchronous Advantage Actor-Critic (A3C) reinforcement learning algorithm, selecting and tuning hyperparameters, and ensuring the correct technical integration of the learning algorithm within the simulation-based optimization architecture.
Shariqa	• Project Manager • Requirements Engineer • Acceptance Manager	Responsible for overall project coordination and scheduling, definition and documentation of stakeholder, functional, and quality requirements, and specification of acceptance criteria for evaluating the A3C-based optimization results in accordance with the defined system objectives.
Revathi Podaralla	• Data Engineer • Test & Validation Engineer • Integration Engineer	Responsible for handling and managing simulation inputs and outputs, supporting reward data processing during A3C training, integrating the optimization software with the simulation environment, and validating optimization results against defined engineering requirements and acceptance criteria.

Stakeholder Definition

This project is developed as a conceptual proof-of-concept for simulation-based optimization using reinforcement learning in an engineering context. The stakeholders identified below represent typical user roles that would interact with or benefit from such a system in an industrial or research-oriented setting. They are not members of the project team but external roles whose interests influence system requirements and acceptance criteria.

Stakeholder 1: Powertrain / Vehicle Development Engineer

Description:

The Powertrain or Vehicle Development Engineer is responsible for defining, evaluating, and refining drivetrain configurations with respect to performance, efficiency, and engineering feasibility. This role is typically involved during early concept and pre-development phases, where simulation-based analysis plays a critical role in narrowing down viable design alternatives before physical prototyping is feasible.

The stakeholder commonly works with parameterized simulation models to assess the influence of drivetrain parameters—such as gear ratios, engine displacement, and compression ratio—on system-level performance indicators.

Operational Context

In industrial practice, the Powertrain Engineer operates under time and cost constraints while facing increasingly complex design spaces. Modern drivetrain systems involve nonlinear interactions and conflicting objectives, making exhaustive manual parameter studies impractical. Simulation-based optimization tools are therefore used to support early decision-making and reduce design uncertainty.

Interest in the Machine Learning Application:

The stakeholder is interested in leveraging the reinforcement learning–based optimization system to automatically explore large, multidimensional design spaces and identify promising parameter configurations. Of particular interest is the ability of the A3C algorithm to learn from interaction with a black-box simulation model and to balance conflicting objectives such as minimizing fuel consumption while maintaining acceptable drivetrain elasticity.

The system is perceived as a decision-support tool rather than a fully autonomous design solution. The engineer remains responsible for interpreting the results and assessing engineering plausibility.

Interaction with the System

The Powertrain Engineer primarily interacts with the system at a conceptual level by:

- Defining feasible parameter ranges and engineering constraints
- Reviewing optimization results and performance trends
- Comparing optimized configurations against baseline designs
- Selecting candidate solutions for further analysis or refinement

The engineer does not directly modify the reinforcement learning algorithm but relies on the system to provide transparent and interpretable outputs.

Expected Benefit:

The main benefit for this stakeholder is a reduction in engineering effort and decision uncertainty during early-stage design. The system supports faster identification of high-performing configurations while ensuring that all evaluated solutions comply with defined engineering constraints and system boundaries.

Stakeholder Goals and Concerns

- Achieve improved fuel efficiency without violating drivetrain constraints
- Maintain interpretability of optimization results for engineering decision-making
- Avoid infeasible or unrealistic design recommendations

Stakeholder 2: Simulation and Optimization Tool Engineer

Description:
The Simulation and Optimization Tool Engineer is responsible for developing, integrating, and maintaining software tools used for simulation-driven engineering analysis. This stakeholder focuses on system robustness, modularity, and technical correctness rather than on domain-specific design decisions.

The role typically bridges simulation models and optimization algorithms, ensuring reliable data exchange and stable execution within automated workflows.

Operational Context
In an industrial or research setting, the Simulation Tool Engineer supports multiple users and optimization studies. The stakeholder must ensure that simulation executions are reproducible, scalable, and fault-tolerant. Parallel execution and automated data handling are particularly relevant when integrating reinforcement learning algorithms that require large numbers of simulation evaluations.

Interest in the Machine Learning Application:
This stakeholder is interested in assessing the technical feasibility of integrating the Asynchronous Advantage Actor-Critic (A3C) reinforcement learning algorithm into an existing simulation environment. Key aspects include:

- Automated command-line interaction with simulation executables
- Handling of numerical input and output data
- Parallel execution using multiple workers
- Stability and reproducibility of learning behavior

The system serves as a reference implementation demonstrating how reinforcement learning can be embedded into engineering workflows.

Interaction with the System
The Simulation and Optimization Tool Engineer interacts with the system by:

- Configuring execution parameters (e.g., number of workers, episodes)
- Monitoring simulation runs and learning progress
- Diagnosing execution errors or unstable behavior
- Exporting and validating result datasets

This stakeholder requires visibility into system behavior and intermediate results to ensure correctness.

Expected Benefit:
The primary benefit is the availability of a structured and documented prototype that demonstrates best practices for coupling reinforcement learning with simulation-based optimization. The system supports experimentation with algorithm configurations and provides insights into scalability and robustness challenges.

Stakeholder Goals and Concerns

- Ensure stable and reproducible optimization runs
- Maintain reliable data exchange between optimizer and simulation
- Detect and diagnose execution or convergence issues efficiently

Stakeholder Summary

The identified stakeholders represent two complementary perspectives within simulation-based engineering optimization. The Powertrain / Vehicle Development Engineer focuses on **decision-making and engineering value**, using the system as a decision-support tool for early-stage design exploration. In contrast, the Simulation and Optimization Tool Engineer focuses on **technical correctness, robustness, and scalability**, ensuring that the optimization process executes reliably and produces reproducible results.

Together, these stakeholders define both the **functional expectations** and **quality constraints** of the system. Their combined interests directly influence the formulation of use cases, stakeholder requirements, and acceptance criteria defined in the subsequent technical specification.

Stakeholder	Primary Interactions with the System	Frequency of Interaction	Primary Output Consumed
Powertrain / Vehicle Development Engineer	Definition of design parameter ranges and constraints; review of optimization results; comparison of candidate configurations	Periodic (per design iteration)	Optimized parameter sets, reward trends, simulation performance metrics
Simulation and Optimization Tool Engineer	Configuration of execution parameters; monitoring of parallel optimization runs; validation and export of result data	Frequent (during execution and testing)	Raw simulation outputs, CSV result files, convergence indicators

Project Timeline Overview

The project was conducted over a fixed and limited time frame of approximately 13 days, from 17 January 2026 to 29 January 2026, as defined by the course schedule for *Machine Learning in Engineering Applications*. Due to the academic nature of the project, the timeline reflects a compressed development cycle, focusing on rapid conceptualization, implementation, and evaluation rather than long-term deployment.

The project timeline was structured to follow a logical progression of activities, starting with project planning and requirements definition, followed by system design, implementation of the reinforcement learning optimization framework, and concluding with validation, analysis, and documentation. Parallel activities were used where possible, particularly during implementation and experimentation phases, to efficiently utilize the available time.

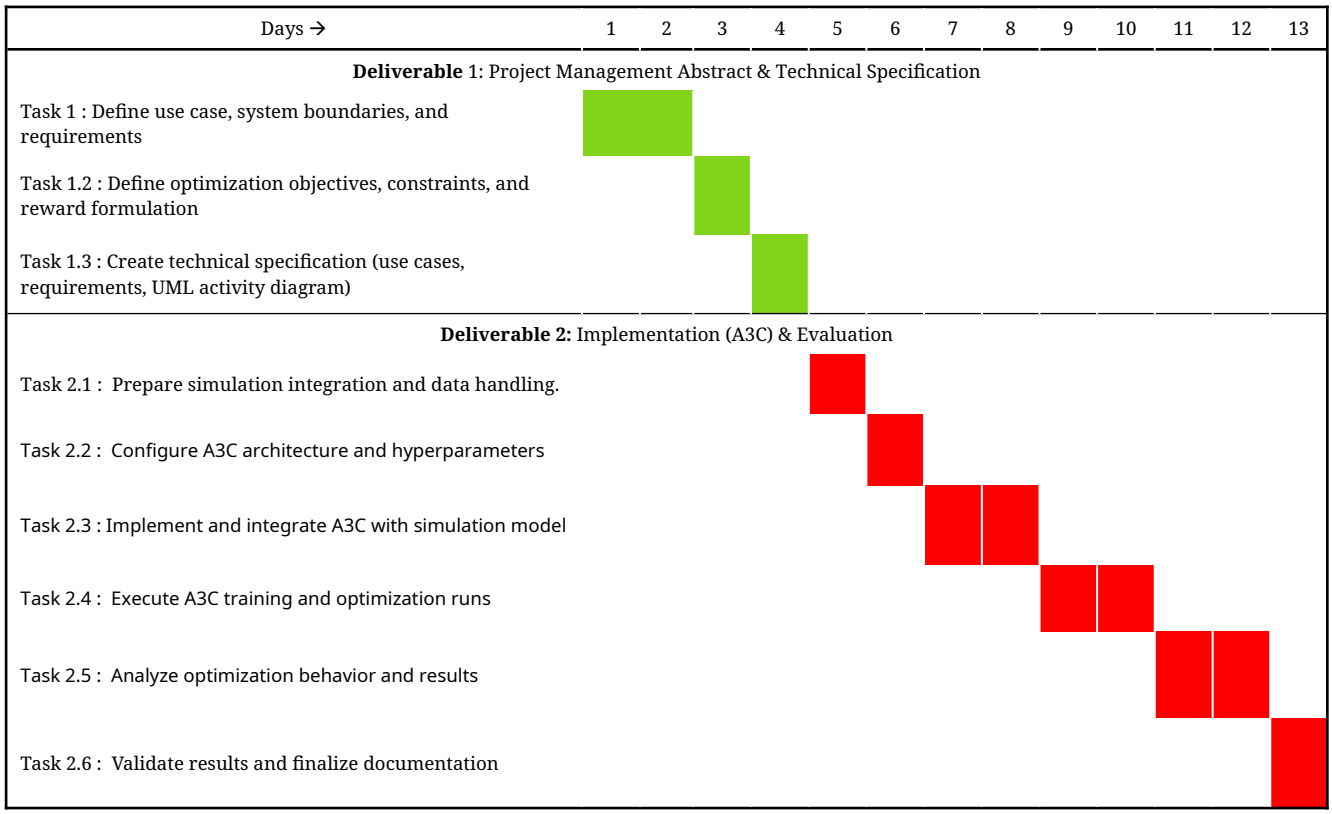
The timeline is visualized using a Gantt chart, which illustrates the sequencing, duration, and overlap of major project phases and milestones.

Item	Description
Project Start Date	17 January 2026
Project End Date	29 January 2026
Total Duration	13 days
Project Type	Academic proof-of-concept
Development Model	Iterative with parallel activities
Primary Constraint	Fixed time frame

Timeline Planning Assumptions

The project timeline was planned under the assumption that a predefined simulation executable and problem domain were available at the start of the project. This allowed the team to focus on reinforcement learning integration, optimization logic, and evaluation rather than on model development. All activities were executed by a small team, requiring parallel task execution and close coordination. No external dependencies or deployment activities were considered, as the project scope was limited to a closed, offline academic prototype.

Gantt-Chart.



Gantt Chart Explanation

The Gantt chart illustrates the planned and executed sequence of project activities across the defined timeline. Overlapping phases indicate parallel work, particularly between implementation, experimentation, and result analysis. Milestones represent clearly defined checkpoints used to assess progress and validate intermediate results. The chart reflects the actual execution of the project and serves as a reference for milestone acceptance and responsibility assignment.

Project Phase	Primary Output
Planning & Requirements	Use cases, stakeholder definitions, requirements
System Design	Technical specification, UML activity diagram
Implementation	A3C optimization framework integrated with simulation
Experimentation	Optimization runs and performance data
Validation & Documentation	Result analysis and finalized deliverables

Milestone Management Approach.

Milestones were defined as objective checkpoints aligned with project phases and deliverables. Each milestone includes measurable acceptance criteria and a single responsible role to ensure clear accountability and transparent progress assessment throughout the project timeline.

Milestone Descriptions.

Abbreviation Gantt-Chart	Short Description Milestone	Acceptance Criteria	Responsible Role (1 role)
M1	Concept & Requirements Defined	The engineering use case, optimization objectives (Fuel vs. Performance), and system constraints are fully documented and approved. The engineering use case, system boundaries, stakeholder, functional, and quality requirements, as well as optimization objectives and constraints are fully documented and reviewed.	Requirements Engineer
M2	A3C Algorithm Concept Finalized	The A3C reinforcement learning approach is selected and justified, including reward formulation, constraint handling strategy, and high-level training workflow.	Machine Learning Engineer
M3	Simulation Integration Operations	The optimization software successfully interfaces with the simulation model. Simulation inputs and outputs are handled correctly, system constraints are enforced, and stable reward data is extracted during execution.	Integration Engineer
M4	A3C Prototype Operational	The A3C implementation executes training runs with parallel workers, updates actor-critic networks correctly, and produces stable reward signals.	Machine Learning Engineer
M5	Optimization Results Validated	Training results are analyzed, reward convergence is demonstrated, and identified design configurations meet defined acceptance criteria and engineering requirements.	Test & Validation Engineer
M6	Final Submission Ready	All deliverables (Project Management & Technical Specification document, source code package, results visualization, and presentation material) are finalized, reviewed, and prepared for submission.	Project Manager

Use Case Descriptions.

Use Case Overview

The following use cases describe the interaction between stakeholders and the simulation-based optimization system. They cover the full life cycle of the application, from execution of reinforcement learning–based optimization, through parallel training, to validation and acceptance of results. The use cases are consistent with the defined stakeholders, requirements, and system workflow.

Use Case-ID - U01

This use case describes the core optimization loop in which the system explores drivetrain design parameters using reinforcement learning while enforcing engineering constraints.

USE CASE 01	Optimize Drivetrain Parameters Using A3C	
Goal in Context	Optimize drivetrain design parameters using reinforcement learning in order to balance fuel consumption and performance-related characteristics under defined system constraints.	
Scope & Level	Simulation-based optimization system Primary Task	
Preconditions	• Valid parameter bounds and system constraints are defined • The optimization system is initialized • The simulation model is available and executable	
Success End Condition	The optimization system generates valid drivetrain parameter configurations that maximize the defined reward function while satisfying all system constraints.	
Failed End Condition	The optimization process terminates without producing valid configurations due to simulation errors or constraint violations.	
Primary, Secondary Actors	**Primary Actor** Powertrain / Vehicle Development Engineer **Secondary Actors** Simulation Model (ConsumptionCar), Optimization System	
Trigger	The engineer starts an optimization run.	
DESCRIPTION	Step	Action
	1	The actor initiates the optimization process.
	2	The system initializes the A3C agent and parallel worker threads.
	3	The system samples normalized design parameters from the current policy.
	4	The sampled parameters are transformed into valid simulation inputs.
	5	The simulation model evaluates the parameters and returns

		performance metrics.
	6	The system computes a reward based on fuel consumption and elasticity values.
	7	The A3C agent updates its policy using the observed reward.
	8	Steps 3–7 are repeated until the maximum number of episodes is reached.
	9	The system stores and outputs high-performing design configurations.
EXTENSIONS	Step	Branching Action
	4a	If sampled parameters violate constraints, the system corrects them before simulation execution.
	5a	If the simulation fails, the system assigns a penalty reward and continues training.
SUB-VARIATIONS		Branching Action
		NONE

Use Case-ID – U02

USE CASE 02	Execute Simulation-Based Training with Parallel Workers
Goal in Context	Execute reinforcement learning training using multiple parallel workers to improve learning efficiency and exploration of the design space.
Scope & Level	Simulation-based optimization system Primary Task
Preconditions	• Simulation model is accessible • System resources for parallel execution are available • A3C algorithm configuration is defined
Success End Condition	Parallel training runs are executed successfully and reward values are collected for policy updates.
Postcondition	Updated global policy parameters are available for subsequent optimization or analysis steps.
Failed End Condition	Training stops prematurely due to system errors or unstable simulation behavior.
Primary, Secondary Actors	**Primary Actor** Machine Learning Engineer

	Secondary Actors Simulation Model, Parallel Worker Threads	
Trigger	The training procedure is started by the engineer.	
DESCRIPTION	Step	Action
	1	The system initializes global and local A3C networks.
	2	Multiple worker threads are started.
	3	Each worker samples actions from the policy.
	4	Workers execute simulations independently.
	5	Reward signals are computed and returned.
	6	Gradients are calculated locally and applied to the global model.
	7	The system aggregates training progress across workers.
	8	Training continues until termination conditions are met.
EXTENSIONS	Step	Branching Action
	3a	If a worker encounters a simulation error, the system assigns a penalty reward.
	6a	If gradient updates fail, the affected worker is restarted.
SUB-VARIATIONS		Branching Action
		NONE

Use Case-ID – U03

USE CASE 03	Analyze and Validate Optimization Results
Goal in Context	Analyze training outcomes and validate optimization results against defined acceptance criteria and engineering requirements.
Scope & Level	Simulation-based optimization system Primary Task
Preconditions	• Valid parameter bounds and system constraints are defined • The optimization system is initialized • The simulation model is available and executable
Success End	Optimization results satisfy acceptance criteria and are approved for

Condition	documentation and submission.	
Failed End Condition	Results do not meet acceptance criteria and require further optimization or parameter tuning.	
Primary, Secondary Actors	**Primary Actor** Test & Validation Engineer. **Secondary Actors** Optimization System, Project Manager.	
Trigger	The validation process is initiated after training completion.	
DESCRIPTION	Step	Action
	1	The system provides recorded reward values and result data.
	2	The actor reviews reward convergence behavior.
	3	High-performing design configurations are identified.
	4	Results are evaluated against defined acceptance criteria.
	5	Validation outcomes are documented and approved.
	6	Validated results are prepared for reporting and documentation.
EXTENSIONS	Step	Branching Action
	4a	If results do not meet acceptance criteria, retraining is recommended.
	5a	If inconsistencies are detected, result analysis is repeated.
SUB-VARIATIONS		Branching Action
		NONE

Use Case Traceability Note

The defined use cases directly serve as the basis for the functional and quality requirements specified in the subsequent requirements section. Each use case is associated with at least one stakeholder requirement and one system-level requirement, ensuring traceability between stakeholder expectations, system behavior, and acceptance criteria.

Technical Specification – Requirements (PARIS Template)

The following requirements are derived directly from the defined use cases and describe stakeholder, functional, and quality aspects of the simulation-based optimization system. All requirements follow the PARIS grammar and are scoped to a conceptual, simulation-based proof-of-concept.

Traceability sentence

Each requirement is traceable to at least one use case and reflects either stakeholder expectations, system functionality, or quality constraints of the simulation-based optimization system.

Stakeholder Requirements

ID	Stakeholder Requirement (PARIS-compliant)
SR-01	When an engineering design optimization task is initiated, the optimization system shall provide simulation-based exploration of drivetrain parameters, so that engineering stakeholders can evaluate design trade-offs without relying on physical prototypes.
SR-02	When conflicting objectives such as fuel consumption and performance characteristics exist, the optimization system shall enable automated evaluation of multiple design objectives, so that engineers can make informed design decisions based on quantitative results.
SR-03	When optimization results are generated, the optimization system shall present high-performing design configurations and training outcomes, so that stakeholders can assess whether the results satisfy defined engineering goals.

Functional Requirements

ID	Functional Requirement (PARIS-compliant)
FR-01	When an optimization episode is executed, the optimization system shall sample normalized design parameters and transform them into valid simulation inputs, so that all evaluated configurations remain within defined system boundaries and constraints.
FR-02	When the simulation model evaluates a candidate design configuration, the optimization system shall compute a reward based on fuel consumption and drivetrain elasticity metrics, so that reinforcement learning updates can be performed consistently.
FR-03	When reinforcement learning training is active, the optimization system shall execute the A3C algorithm using asynchronous parallel worker processes, so that learning efficiency and exploration of the design space are improved.

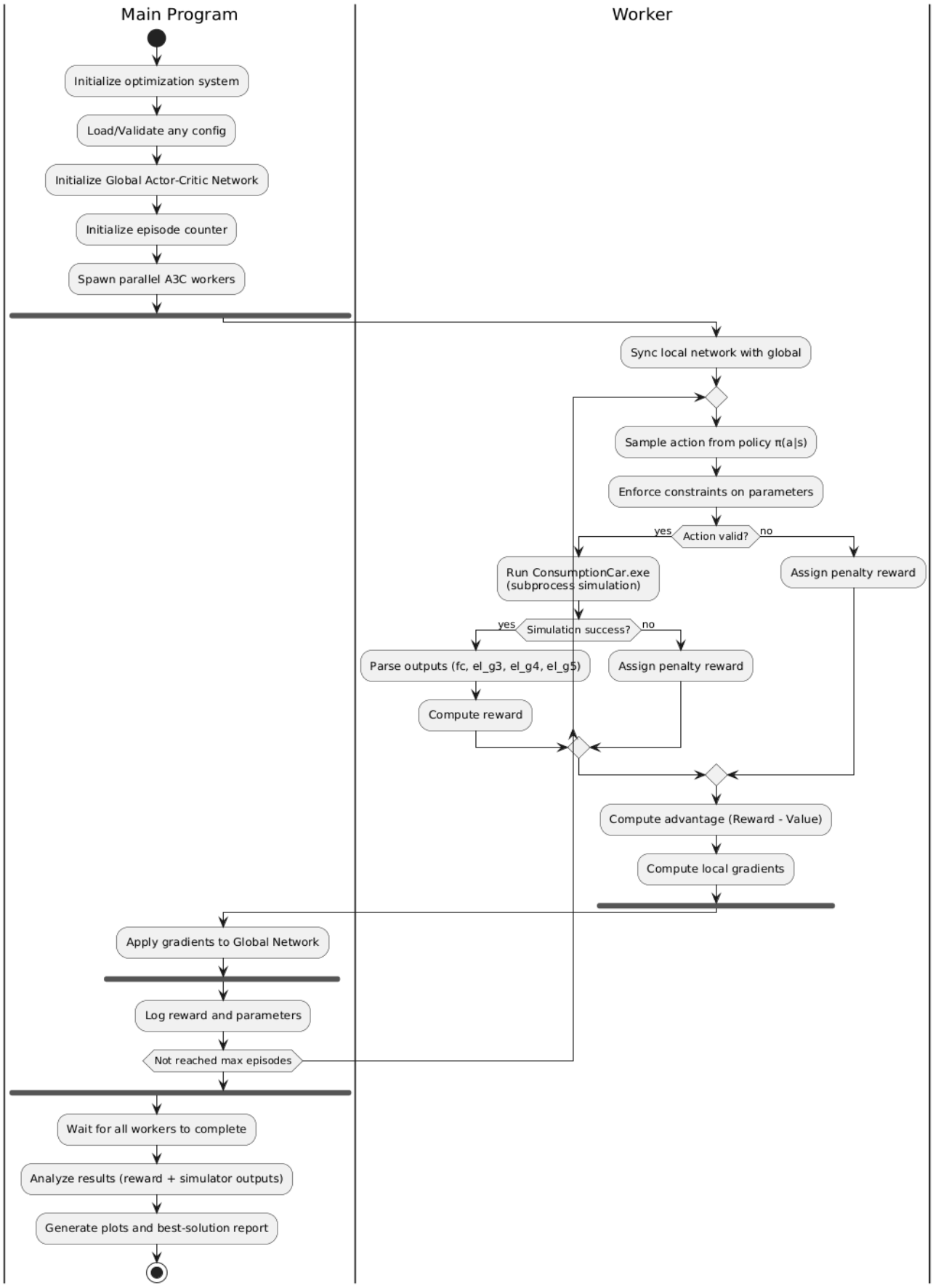
Quality Requirements

ID	Quality Requirement (PARIS-compliant)
QR-01	**When** simulation-based training is executed, **the optimization system shall** handle simulation errors by assigning penalty rewards and continuing execution, **so that** training robustness is maintained despite occasional simulation failures.
QR-02	**When** optimization results are generated, **the optimization system shall** ensure that only constraint-compliant design configurations are considered valid, **so that** all reported results are engineering-feasible.
QR-03	When reinforcement learning training is completed, the optimization system shall provide reproducible and documented result data and reward trends, so that optimization behavior can be analyzed and validated reliably.

Requirements Traceability Matrix.

Requirement ID	Requirement Type	Source Use Case(s)
SR-01	Stakeholder	UC-01
SR-02	Stakeholder	UC-01
SR-03	Stakeholder	UC-03
FR-01	Functional	UC-01
FR-02	Functional	UC-01
FR-03	Functional	UC-02
QR-01	Quality	UC-01, UC-02
QR-02	Quality	UC-01
QR-03	Quality	UC-03

UML Diagram


Main Program
Worker
Initialize optimization system
Load/Validate any config
Initialize Global Actor-Critic Network
Initialize episode counter
Spawn parallel A3C workers
Sync local network with global
Sample action from policy π(a|s)
Enforce constraints on parameters
Action valid?
yes
no
Run ConsumptionCar.exe (subprocess simulation)
Assign penalty reward
Simulation success?
yes
no
Parse outputs (fc, el_g3, el_g4, el_g5)
Assign penalty reward
Compute reward
Compute advantage (Reward - Value)
Compute local gradients
Apply gradients to Global Network
Log reward and parameters
Not reached max episodes
Wait for all workers to complete
Analyze results (reward + simulator outputs)
Generate plots and best-solution report